

test1_review

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:53pm

Quiz Instructions

3 attempts/ The grade does not count. grade is out of 1.2

The review test does not have to reflect toally the true exam. Review slides and HW as well

See tutorial in dropbox



Question 1 0.1 pts

A stone is thrown horizontally with an initial speed of 10 m/s from the edge of a cliff. A stopwatch measures the stone's trajectory time from the top of the cliff to the bottom to be 4.3 s. What is the height of the cliff if air resistance is negligibly small?

hint: ignore the fact that the stone wa thrown horizontally. Along the vertical, it is in free-fall and there is no initial velocity (along the veritcal). So its a free-fall problem. you are solving for the vertical displacement delta Y

☐

43 m

☐

22 m

☐

91 m

☐

77 m



Question 2 0.1 pts

A laser is thrown upward with a speed of 12 m/s on the surface of planet X where the acceleration due to gravity is 1.5 m/s^2 and there is no atmosphere. What is the maximum height reached by the laser?

☐

8.0 m

☐

48 m



18 m



144 m



Question 3 0.1 pts

A laser is thrown upward with a speed of 12 m/s on the surface of planet X where the acceleration due to gravity is 1.5 m/s^2 and there is no atmosphere. How long does it take for the laser to reach the maximum height?



8.0 s



14 s



11 s



16 s



Question 4 0.1 pts

A jet plane is launched from a catapult on an aircraft carrier. In 2.0 s it reaches a speed of 42 m/s at the end of the catapult. Assuming the acceleration is constant, how far did it travel during those 2.0 s?



24 m



16 m



84 m



42 m



Question 5 0.1 pts

A car initially traveling at 60 km/h accelerates at a constant rate of 2.0 m/s^2 . How much time is required for the car to reach a speed of 90 km/h?



30 s



4.2 s



45 s



15 s



Question 6 0.1 pts

An airplane increases its speed at the average rate of 15 m/s^2 . How much time does it take to increase its speed from 100 m/s to 160 m/s ?



4.0 s



0.25 s



17 s

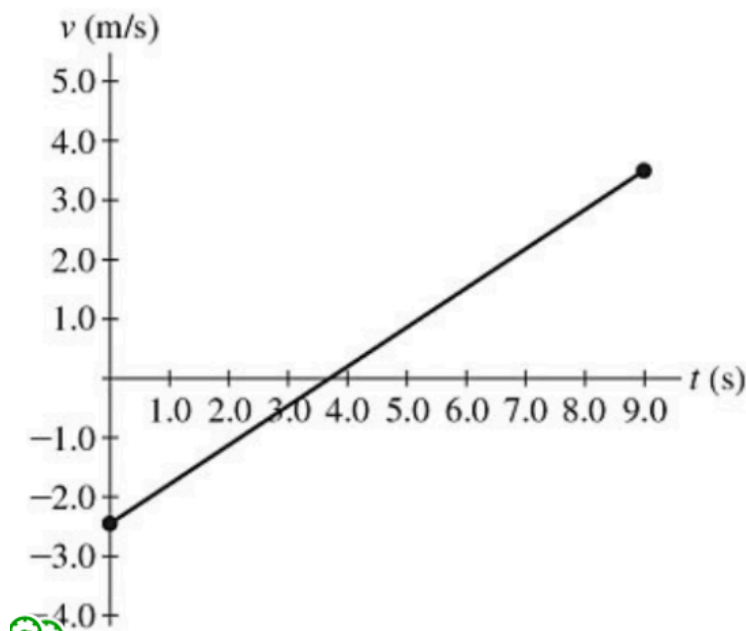


0.058 s



Question 7 0.1 pts

The motion of a particle is described in the velocity vs. time graph shown in the figure.



Over the nine-second interval shown, we can say that the *speed* of the particle



increases and then decreases.



remains constant.



only decreases.



only increases.

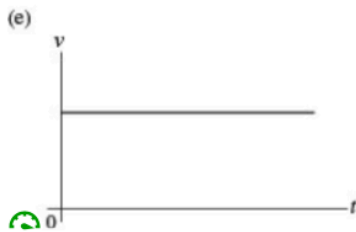
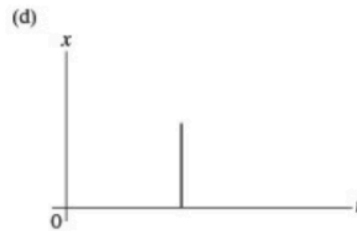
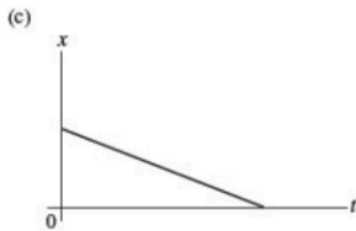
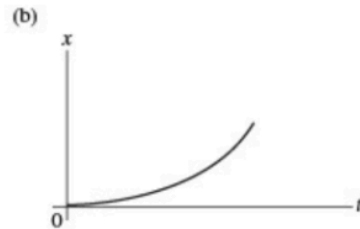
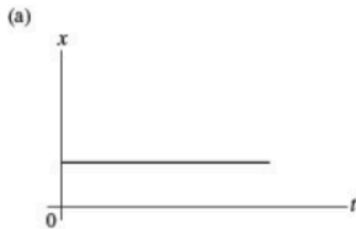


decreases and then increases.



Question 8 0.1 pts

Which of the following graphs represent an object at rest? (There could be more than one correct choice.)



graph e



graph a



graph b



graph c



graph d



Question 9 0.1 pts

Suppose that a car traveling to the east (+x direction) begins to slow down as it approaches a traffic light. Which statement concerning its acceleration must be correct?



Its acceleration is in the $+x$ direction.



Its acceleration is zero.



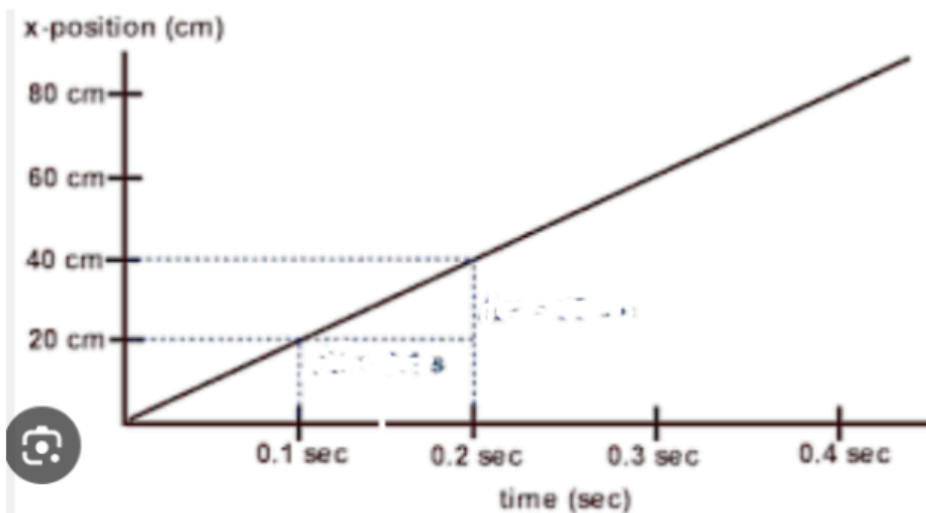
Its acceleration is in the $-x$ direction.



Its acceleration is decreasing in magnitude as the car slows down.



Question 10 0.1 pts



Here is a graph position (number line) of a toy car versus time. The velocity of the car (convert to right units)



2m/s



4m/s



1m/s



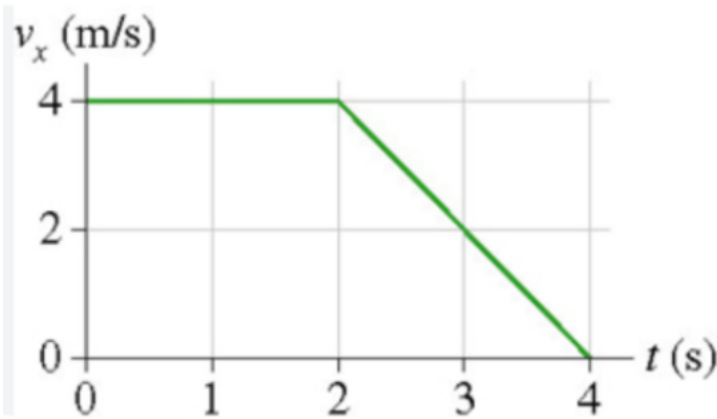
200m/s



Not enough information



Question 11 0.1 pts



During $t=2$ s and $t=4$ s what is the displacement of the toy car ? (hide the hint first)

hint: what is the initial velocity at $t=2$ s? (read the graph)

What is the acceleration between 2s and 4s? (its negative. the acceleration is the ... of $V(t)$)

what is the time elapsed ?

Now you can find the displacement

- ☐ 8 m
- ☐ 4m
- ☐ 12 m
- ☐ 2 m



Question 12 0.1 pts

The distance to the Andromeda Galaxy is estimated at about 2×10^6 light years. A light year is the distance traveled by light in one year; if the speed of light is 3×10^8 m/s, about how far is it from our galaxy to Andromeda? (1 year = 3.15×10^7 s)

- ☐ 1×10^{18} m
- ☐ 2×10^{22} m
- ☐ IDK
- ☐ 6×10^{12} m
- ☐ 10×10^{15} m

Not saved

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